

LOCALIZATION IN THE COMMUTANT, STAGE 9:

the mean of a random ReLU network as a central localization,
with the exact slice transport, the collective-mode sensitivity law
and the capacity of input tilts

Stage 9 of the quantum-cut / noncommutative-geometry programme

9 October 2026

Abstract

We apply the stage-8 principle (a localization process is a positive martingale in the commutant, whose central part is the transverse measure on the leaf space) to the WhestBench problem: the per-neuron mean $\mathbb{E} h_L$ of a bias-free He-initialised ReLU multilayer perceptron with standard Gaussian input, $n = 1024$, $L = 16$. The cumulant chains that dominate the leaderboard are read as localizations: the births of non-Gaussianity at the ReLU kinks are the linear-tilt responses of the gate algebra, the chain is the trickle-down expansion of that algebra's product state, and the one object that no first-order chain carries correctly is the *central* variable—the collective activity mode, a scale mixture—which is exactly the leaf-space part of the localization. Main results. (A) Input-side linear tilts have localization capacity $O(k/n)$ for the born non-Gaussianity and cost $\geq 2k + 1$ chain evaluations: input localization is useless, which forces the neuron-side (gate) decomposition. (B) A sensitivity law: the output mean responds to a relative error ε of the collective mode's variance with error $c\varepsilon$ per layer, $c \approx 0.3$, so the collective mode must be accurate to 10^{-3} per layer while the bulk may be off by 10^{-1} ; on the official networks the chain's covariance is accurate everywhere except in that one direction. (C) The exact first-order transport of the $(2, 1)$ slices and of the diagonal through ReLU, in closed form, including the transposed-slice and the $\kappa_4 \rightarrow \kappa_3$ feeds, and the exact-by-homogeneity transport of the scale mixture. (D) On the official networks, the terms prescribed by the central/Cartan split account for the measured error ladder $8.5 \cdot 10^{-7} \rightarrow 5.7 \cdot 10^{-7}$ of a clean chain and locate the remaining deficit in the transported third-cumulant slices (25% low at depth). Every identity is checked by the accompanying script.

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1 The problem and the principle

1.1 The estimation problem

Fix weights $W_1, \dots, W_L \in \mathbb{R}^{n \times n}$, drawn once from $N(0, 2/n)$ (He initialisation) and then held fixed. For an input $x \in \mathbb{R}^n$ put $h_0 = x$, $z_l = W_l h_{l-1}$, $h_l = \text{relu}(z_l)$. The target is the vector of post-activation means

$$u = \mathbb{E}_{x \sim N(0, I_n)} h_L(x) \in \mathbb{R}^n,$$

scored by the mean squared error over the n neurons, multiplied by $\max(0.1, C/B)$ with C the metered floating point count and $B = 2^{41}$. The map $F = h_L$ is positively homogeneous of degree one and piecewise linear. The reference answer is a Monte Carlo average over 10^9 inputs, whose per-neuron noise ($\sim 10^{-5}$) is below every error discussed here.

1.2 Estimators are functions on the state space; exact ones are affine

Let $\mathcal{S}$ be the set of probability laws ν on $\mathbb{R}^n$ with enough moments. Every deterministic estimator we consider is a map $E : \mathcal{S}_0 \rightarrow \mathbb{R}^n$ defined on a subfamily $\mathcal{S}_0 \subseteq \mathcal{S}$ that contains the point masses δ_x and the Gaussians $N(m, \Sigma)$; a cumulant chain is defined on Gaussians by running its recursion from the given (m, Σ) , and on point masses by the forward pass, $E(\delta_x) = F(x)$. The exact answer

$$u(\nu) = \nu(F) = \int F d\nu$$

is *affine* on $\mathcal{S}$. This is the whole content of exactness: in the commutant language of [1], u is the pairing of the state ν with the observable F , and a localization process of ν (a positive martingale of Radon–Nikodym derivatives in the commutant) turns u into a martingale. A chain is not affine: it is exact on point masses and wrong on mixtures, and the gap between the two is what we have to compute.

Definition 1.1 (Localization process). A localization process of ν is a sequence or continuous family $(\nu_t)_{t \geq 0}$ of random laws with $\nu_0 = \nu$, adapted to a filtration $(\mathcal{F}_t)$, such that $\mathbb{E}[\nu_{t+s}(g) \mid \mathcal{F}_t] = \nu_t(g)$ for every bounded g , and *complete* if $\nu_t \rightarrow \delta_X$ almost surely for a random $X \sim \nu$ [4, 1]. Three schemes are used below: Eldan’s isotropic tilt, $\nu_t = \text{Law}(x \mid y_t)$ with $y_t = tX + B_t$, so that for $\nu = N(0, I)$ one has $\nu_t = N(m_t, I/(1+t))$ with m_t a martingale of quadratic variation $d\langle m \rangle_t = (1+t)^{-2}I dt$; the *directional* tilt, the same observation with $y_t = t\langle v, X \rangle + B_t$ for a unit vector v , which leaves $\nu_t = N(m_t, I - \frac{t}{1+t}vv^\top)$; and pinning, which reveals the coordinates of X one at a time.

Theorem 1.2 (Defect–path formula). *Let $E : \mathcal{S}_0 \rightarrow \mathbb{R}^n$ be twice continuously Fréchet differentiable along the Gaussian family and exact on point masses, and let (ν_t) be a complete localization process of $\nu \in \mathcal{S}_0$ whose components stay in $\mathcal{S}_0$ and have bounded quadratic variation. Then*

$$u(\nu) - E(\nu) = -\mathbb{E} \int_0^\infty \frac{1}{2} E''(\nu_t) [d\langle \nu \rangle_t] = -\mathbb{E} \int_0^\infty \text{drift}_t(E(\nu_t)) dt. \quad (1)$$

In words: the error of the estimator at ν is minus the expected total drift of the estimator along any complete localization of ν , and that drift is the second variation of E in the directions the scheme localizes.

Proof. $u(\nu_t)$ is a martingale (the law of iterated expectations, or affinity of u plus the martingale property of ν_t), so $u(\nu) = \mathbb{E} u(\nu_\infty) = \mathbb{E} F(X)$. By Itô’s formula for the semimartingale ν_t in the parametrisation of $\mathcal{S}_0$, $dE(\nu_t) = E'(\nu_t)[d\nu_t] + \frac{1}{2} E''(\nu_t)[d\langle \nu \rangle_t]$, and the first term has zero mean. Hence $\mathbb{E} E(\nu_\infty) - E(\nu) = \mathbb{E} \int_0^\infty \frac{1}{2} E''(\nu_t)[d\langle \nu \rangle_t]$. Exactness on point masses gives $E(\nu_\infty) = F(X)$ almost surely, so $\mathbb{E} E(\nu_\infty) = u(\nu)$. For a discrete scheme the same identity holds with the drift replaced by the conditional Jensen gaps $\mathbb{E}[E(\nu_{t+1}) \mid \mathcal{F}_t] - E(\nu_t)$. $\square$

Corollary 1.3 (Eldan’s path for a homogeneous network). *Write $e(y) = E(N(y, I))$ for a chain E that is exact on point masses and transforms like F under scaling, $E(N(m, sI)) = \sqrt{s} e(m/\sqrt{s})$. Then with $\delta(y) = e(y) - y \cdot \nabla e(y) - \Delta e(y)$,*

$$u - e(0) = -\frac{1}{2} \int_0^\infty (1+\tau)^{-3/2} \mathbb{E}_{y \sim N(0, \tau I)} \delta(y) d\tau, \quad (2)$$

and the weight $\frac{1}{2}(1+\tau)^{-3/2}d\tau$ is a probability measure on $[0, \infty)$. At the origin $\delta(0) = e(0) - \Delta e(0)$, and the exact answer satisfies $u = \Delta_y u(y)|_{y=0}$ (Euler–Stein).

Proof. On Eldan’s path $\nu_t = N(m_t, s_t I)$, $s_t = 1/(1+t)$, $ds_t = -s_t^2 dt$, $d\langle m \rangle_t = s_t^2 I dt$. With $E(m, sI) = \sqrt{s} e(m/\sqrt{s})$ one finds $\partial_s E = \frac{1}{2\sqrt{s}}(e - y \cdot \nabla e)$ and $\frac{1}{2} \Delta_m E = \frac{1}{2\sqrt{s}} \Delta e$ at $y = m/\sqrt{s}$, so the drift is $-s^2(\partial_s E - \frac{1}{2} \Delta_m E) = -\frac{1}{2} s^{3/2} \delta(y)$. Since $y_t = m_t \sqrt{1+t} \sim N(0, tI)$, Theorem 1.2 gives (2) with $\tau = t$. For the exact answer, $u(y) = \mathbb{E} F(y+Z)$ obeys the heat equation $\partial_s u = \frac{1}{2} \Delta_m u$, and for a 1-homogeneous F Euler’s relation $x \cdot \nabla F = F$ and Stein’s lemma $\mathbb{E}[x \cdot \nabla F(x)] = \mathbb{E} \Delta F(x)$ give $u = \Delta_y u|_0$. $\square$

This corollary is the heat-flow defect of the parallel note (its “heat defect” is $D = \partial_\Sigma E - \frac{1}{2} \nabla_m^2 E$; here $\frac{1}{2} s^{3/2} \delta$ is the trace of $D \Sigma_t^2$ on the isotropic path). Theorem 1.2 is the scheme-independent statement behind it, and the choice of scheme is where the stage-8 picture adds something.

Corollary 1.4 (Directional and pinning schemes). *(i) For the directional tilt along a unit vector v , the path is $\nu_t = N(m_t, I - \frac{t}{1+t}vv^\top)$ with $d\langle m \rangle_t = (1+t)^{-2}vv^\top dt$, the endpoint is $\nu_\infty = N(\langle v, X \rangle v, I - vv^\top)$, and*

$$\mathbb{E}_y E(N(yv, I - vv^\top)) - E(N(0, I)) = -\int_0^\infty (1+t)^{-2} \mathbb{E} \left[\frac{1}{2} \partial_v^2 E - \partial_{\Sigma_{vv}} E \right] (\nu_t) dt,$$

the left side being the error of the v -localized estimator minus the error of E . So the drift along v measures the difference between the chain and its v -localization, and the chain's own error is recovered from directional drifts only by summing over an orthonormal basis of directions—which is Eldan's path. (ii) For pinning (reveal X_i in order), the Jensen gap at step i is $\mathbb{E}_{X_i} E(\nu_{i-1} | X_i) - E(\nu_{i-1})$, a one-dimensional Gaussian integral of the chain run on the conditioned law; the chain's error is minus the sum of these n gaps.

Remark 1.5 (Which errors a scheme sees). The second variation $E''(\nu)[\dot{\nu}, \dot{\nu}]$ is a quadratic form on the tangent space of $\mathcal{S}_0$ at ν . A scheme supplies a positive form $d\langle \nu \rangle$ (its driving covariance C_t in the language of [1, Thm. 3.3], which also fixes the trickle-down term KC_tK of that theorem). Eldan's path drives every mean direction equally and the covariance only through the scale; the directional tilt drives one direction; pinning drives the coordinate directions and their conditional variances. Which scheme probes a given chain cheaply is therefore a question about the *spectrum* of the chain's second variation on the input side. Section 3 measures it: for the chains of this note the excess second variation $\partial_v^2 e - \Delta e/n$ over unit input directions v has a flat spectrum (the top direction carries 5% of it at $n = 256$, and is not the coherent direction $W_1^\top \mathbf{1}$), and single directional second differences are nearly uncorrelated with the error. The chain's error is an isotropic, trace-like object on the input side—the Laplacian of Corollary 1.3—and the only cheap way to get at it is not a tilt of the input but a second-order expansion *inside* the chain (Section 4). The collective mode that the chain mishandles lives in the gate algebra of the hidden layers, not in the input, which is the content of the capacity bound of Section 3.

2 Births are linear-tilt responses of the gate algebra

We now compute the second variation of a chain explicitly. The result is that the objects a cumulant chain transports (births at the kinks, gated arms, $(2, 1)$ slices) are the first and second derivatives of the Gaussian moment map under a shift of the mean, and that the Euler–Stein defect of Section 3 is the curvature of the downstream chain contracted with the Gram matrix of those derivatives.

2.1 The chaos index shifts under a tilt

For $u \sim N(0, 1)$ and $a \in \mathbb{R}$ let $r_a(u) = \text{relu}(u+a)$ with Hermite coefficients $d_k(a) = \mathbb{E}[r_a(u)\text{He}_k(u)]$:

$$d_0 = \varphi(a) + a\Phi(a), \quad d_1 = \Phi(a), \quad d_k = \varphi(a)\text{He}_{k-2}(-a) \quad (k \geq 2),$$

so that for standardised (u, v) with correlation ρ , $\mathbb{E}[r_a(u)r_b(v)] = \sum_{k \geq 0} \rho^k d_k(a)d_k(b)/k!$.

Lemma 2.1 (Index shift). $d'_k(a) = d_{k+1}(a)$ for every $k \geq 0$.

Proof. $d'_0 = -a\varphi + \Phi + a\varphi = \Phi = d_1$ and $d'_1 = \varphi = d_2$. For $k \geq 2$, with $x = -a$, $\frac{d}{da}[\varphi(a)\text{He}_{k-2}(x)] = \varphi[x\text{He}_{k-2}(x) - (k-2)\text{He}_{k-3}(x)] = \varphi\text{He}_{k-1}(x) = d_{k+1}(a)$ by $\text{He}'_m = m\text{He}_{m-1}$ and the three-term recursion $\text{He}_{m+1}(x) = x\text{He}_m(x) - m\text{He}_{m-1}(x)$. $\square$

(Checked numerically to $2 \cdot 10^{-8}$ for $k \leq 10$ in `verify_stage9.py`.) The lemma is Stein's identity $\mathbb{E}[f'(u)\text{He}_k(u)] = \mathbb{E}[f(u)\text{He}_{k+1}(u)]$ for $f = r_a$; what matters here is that the tilt $a \mapsto a + t$ of the Gaussian acts on the chaos expansion of the RELU by raising the index.

Let now $z \sim N(m, S)$ be the pre-activation of one layer, $\sigma_a = \sqrt{S_{aa}}$, $\alpha_a = m_a/\sigma_a$, $\rho_{ab} = S_{ab}/\sigma_a\sigma_b$, and

$$\mu_a = \sigma_a d_0(\alpha_a), \quad M_{ab} = \mathbb{E}[\text{relu } z_a \text{ relu } z_b] = \sigma_a \sigma_b \sum_k \rho_{ab}^k d_k(\alpha_a) d_k(\alpha_b) / k!$$

the post-activation mean and raw second moment. Lemma 2.1 gives every derivative of the closure:

$$\partial_{m_a} \mu_a = \Phi(\alpha_a) =: g_a, \quad \partial_{m_a}^2 \mu_a = \varphi(\alpha_a) / \sigma_a =: w_a, \quad (3)$$

$$\partial_{m_a} M_{ab} = \sigma_b \sum_k \rho^k d_{k+1}(\alpha_a) d_k(\alpha_b) / k! = \mathbb{E}[\theta(z_a) \text{relu } z_b] =: T_{ab}, \quad (4)$$

$$\partial_{S_{ab}} M_{ab} = \sum_k \rho^k d_{k+1}(\alpha_a) d_{k+1}(\alpha_b) / k! = \mathbb{E}[\theta(z_a) \theta(z_b)] = \partial_{m_a} \partial_{m_b} M_{ab} =: G_{ab}, \quad (5)$$

$$\partial_{m_a}^2 M_{ab} = \frac{\sigma_b}{\sigma_a} \sum_k \rho^k d_{k+2}(\alpha_a) d_k(\alpha_b) / k! = w_a \mu_b + \frac{\sigma_b}{\sigma_a} \sum_{k \geq 1} \rho^k d_{k+2}(\alpha_a) d_k(\alpha_b) / k!. \quad (6)$$

Here θ is the Heaviside gate. The identities with expectations are Price's theorem $\partial_{S_{ab}} \mathbb{E}[f(z_a)h(z_b)] = \mathbb{E}[f'(z_a)h'(z_b)]$ and Stein's lemma; the closure is exact for one layer because it satisfies the heat equation $\partial_{S_{ab}} = \partial_{m_a} \partial_{m_b}$ in its own parameters. The dictionary with the cumulant chain of [2] and of Section 5 is now literal: $w_a = \varphi(\alpha_a)/\sigma_a$ is the *birth* strength at the kink of neuron a , the second derivative of the mean under a tilt; $g_a = \Phi(\alpha_a)$ gates the arms; T_{ab} is the exact $(2, 1)$ slice function $\mathbb{E}[\theta(z_a) \text{relu } z_b]$ through which a tilt at a moves the covariance row of a ; G_{ab} is the gate covariance $\mathbb{E}[\theta_a \theta_b]$ that transports a covariance perturbation. Births are the curvature of the tilt response; arms and slices are its gradient.

2.2 The defect is downstream curvature against the tilt Gram

Write the chain as $e(y) = \mathcal{G}(\theta(y))$, where $\theta(y) = (\mu(y), M(y))$ are the exact first-layer post-activation moments for the input law $N(y, I)$ (exact because $z_1 = W_1 x$ is Gaussian) and $\mathcal{G}$ is the chain from layer 2 on, a function of the Gaussian surrogate $N(\mu, M - \mu\mu^\top)$ of h_1 .

Theorem 2.2 (Defect = curvature $\times$ Gram). *Let $\mathcal{G}$ be homogeneous, $\mathcal{G}(\lambda\mu, \lambda^2 M) = \lambda \mathcal{G}(\mu, M)$, and twice differentiable. Then for every y ,*

$$\delta(y) = e - y \cdot \nabla e - \Delta e = - \sum_{A,B} \partial_A \partial_B \mathcal{G}(\theta(y)) \Gamma_{AB}(y), \quad \Gamma_{AB}(y) = \langle \nabla_y \theta_A(y), \nabla_y \theta_B(y) \rangle, \quad (7)$$

the sum running over the moment indices $A, B \in \{\mu_a\} \cup \{M_{ab}\}$. With $S = W_1 W_1^\top$ and the rows w_a of W_1 , at $y = 0$ the Gram matrix is

$$\Gamma_{\mu_a \mu_b} = g_a g_b S_{ab}, \quad \Gamma_{\mu_a M_{bc}} = g_a (T_{bc} S_{ab} + T_{cb} S_{ac}), \quad \Gamma_{M_{ab} M_{cd}} = T_{ab} T_{cd} S_{ac} + T_{ab} T_{dc} S_{ad} + T_{ba} T_{cd} S_{bc} + T_{ba} T_{dc} S_{bd}.$$

Consequently $u - e(0) = \int_0^\infty w(\tau) \mathbb{E}_{y \sim N(0, \tau I)} \langle \nabla^2 \mathcal{G}(\theta(y)), \Gamma(y) \rangle d\tau$ with $w(\tau) = \frac{1}{2}(1 + \tau)^{-3/2}$.

Proof. Each $\theta_A(y) = \mathbb{E} g_A(y + Z)$, $Z \sim N(0, I)$, with g_A positively homogeneous of degree $p_A \in \{1, 2\}$, satisfies the heat equation $\partial_s \theta_A = \frac{1}{2} \Delta \theta_A$ along $N(y, sI)$ and the scaling $\theta_A(y, s) = s^{p_A/2} \theta_A(y/\sqrt{s}, 1)$, whence $p_A \theta_A - y \cdot \nabla \theta_A - \Delta \theta_A = 0$. By the chain rule $\nabla e = \sum_A \partial_A \mathcal{G} \nabla \theta_A$ and $\Delta e = \sum_A \partial_A \mathcal{G} \Delta \theta_A + \sum_{A,B} \partial_A \partial_B \mathcal{G} \Gamma_{AB}$, so $\delta = \mathcal{G} - \sum_A p_A \theta_A \partial_A \mathcal{G} - \sum_{AB} \partial_A \partial_B \mathcal{G} \Gamma_{AB}$, and Euler's relation for the weighted homogeneity of $\mathcal{G}$ kills the first two terms. The Gram entries follow from (3)–(4): $\nabla_y \mu_a = g_a w_a$ and $\nabla_y M_{ab} = T_{ab} w_a + T_{ba} w_b$. The last statement is Corollary 1.3. $\square$

Three remarks. (1) The theorem is exact and holds for every homogeneous chain (Gaussian closure or any cumulant chain run from the Gaussian surrogate of h_1); nothing about the chain's internals enters except its curvature in the first-layer moments. (2) The Gram matrix has the *CP-leg* form of the cumulant chain: a gated covariance $g \circ S \circ g$ (the arms), slice products $T \circ S$ (the $(2, 1)$ legs), and the four-term (M, M) block in which the pair (a, b) is linked to (c, d) through a single covariance entry. The same construction at a hidden layer l (tilt the hidden mean m_l with the chain's C_l in place of I) gives the error of the chain started from the surrogate at layer l , Err_l , and the differences $\text{Err}_{l-1} - \text{Err}_l$ are the injected errors of Section 7. (3) The contraction $\langle \nabla^2 \mathcal{G}, \Gamma \rangle$ can be evaluated inside the chain by propagating the second-order response of the closure to a Gaussian shift of the mean, i.e. the covariances $\mathbb{E}[\delta m \delta m^\top]$, $\mathbb{E}[\delta m_a \delta S_{ab}]$, $\mathbb{E}[\delta m_a \delta S_{bb}]$, $\mathbb{E}[\delta S_{aa} \delta S_{bb}]$, $\mathbb{E}[\delta S_{ab} \delta S_{aa}]$ of the perturbation; every one of them closes at $O(n^3)$ per layer through (3)–(6) except $\mathbb{E}[\delta S_{ab}^2]$, the variance of the transported cross-covariance, whose exact propagation is $O(n^4)$. This is the same obstruction that forces a cumulant chain to carry slices of κ_3 rather than the tensor, met here from the other side: the second variation of a chain has the complexity of the chain's own closure error.

2.3 The organisers' chain in this language

The reference implementation of [2] (`alignment-research-center/mlp_cumulant_propagation`, which we ported to numpy and matched to 10^{-15} on a small network, `whest/kprop3.py`) is the first-order theory of this section written as a term table. Its *Wick coefficients* $\mathbb{E}[\partial^k \text{relu}(Z)^p]$ are the index shifts of Lemma 2.1: for $p = 1$ they are $\sigma^{1-k} d_k(\alpha)$, and for $p = 2, 3, 4$ the Hermite coefficients of relu^p (`relu2_hermite`, `relu3_hermite`). Its nonlinear step at $k_{\max} = 3$ sums 36 diagrams (`notes/stage9/terms_k3_simple.tex`): for each output slice (powers p_i of the distinct indices) a product of cumulant blocks of z against the Wick coefficients $\mathbb{E}[\partial^{k_i} \text{relu}^{p_i}]$ with k_i the multiplicity of index i over the blocks, cut off by the power counting $\sum_{\text{blocks}} (|v| - 1 - [v \text{ even}]) \leq k_{\max} - 1$. Read through Theorem 2.2, every term is a second- or third-order tilt response: the gate $\Phi_i \Phi_j$ on C_{ij} is the gradient block $\Gamma_{\mu\mu}$, the birth w_j with two arms $C_{ij} C_{jk}$ is the star of the cumulant chain and the diagonal of the Hessian block Γ_{MM} , and the coefficient $\frac{1}{2} \Phi_i w_j$ of $\kappa_3(z_i, z_i, z_j)$ in $\text{Cov}(h_i, h_j)$ is (9). Two features of the reference matter for the system. First, in simple mode the slice gates are taken at $\rho = 0$ and the pair moment is cut at $O(\rho^2)$: the ρ -dependence of a gate times a cumulant is a cost-3 diagram, dropped by the power counting; our chain's full Hermite series of the Gaussian part is harmless but not what distinguishes the two. Second, the fourth cumulant is kept only as its radial (isotropic) part, $\kappa_4(h) = c \text{Sym}(I \otimes I)$ with one scalar $c = \frac{3}{n(n+2)} (\sum_{a \neq b} \kappa_4(h_a, h_a, h_b, h_b) + \sum_a \kappa_4(h_a))$ per layer, transported as $c \text{Sym}(S \otimes S)$ with $S = WW^\top$: this is the scale mixture of Lemma 4.1, the central variable, and nothing else of κ_4 . The reference thus confirms the split of Section 4 from the other side: the one piece of κ_4 worth carrying is the collective scale.

3 Capacity of input tilts

The stage-8 programme asks first which localization sees the error. Here we settle it for the input side, with two measurements on a small network ($n = 256$, $L = 4$, Monte Carlo truth from 10^7 samples, noise $2 \cdot 10^{-4}$) and one on the official network 0 ($n = 1024$, $L = 16$).

3.1 The Laplacian is the error; its directional parts are not

Table 1 gives, for the Gaussian closure and for the chain of this note (k3v3), the per-neuron correlation between the error $e(0) - u$ and the Euler–Stein defect $\delta(0) = e(0) - \Delta e(0)$ of Corollary 1.3, the least-squares coefficient a in $e(0) - u \approx a \delta(0)$, and the mean squared error after the correction $e(0) - a\delta(0)$. The Laplacian was computed exactly by $2n + 1$ chain evaluations. Two facts. First,

chain	MSE	$ \delta _{\text{rms}}$	$\text{corr}(e - u, \delta)$	a	fraction explained	MSE after $e - a\delta$
Gaussian closure	$1.16 \cdot 10^{-5}$	$9.7 \cdot 10^{-3}$	0.967	0.339	0.936	$7.5 \cdot 10^{-7}$
k3v3	$1.09 \cdot 10^{-6}$	$6.1 \cdot 10^{-3}$	0.835	0.143	0.690	$3.4 \cdot 10^{-7}$

Table 1: Euler–Stein defect against the true error, $n = 256$, $L = 4$ (`scripts/defect_small.py`).

$a < 1$: Corollary 1.3 integrates $\delta(y)$ against $y \sim N(0, \tau I)$ with the weight $\frac{1}{2}(1 + \tau)^{-3/2}$, and $\delta(y)$ decays as $|y|$ grows (a shifted input makes the network more nearly deterministic relative to its fluctuations), so $\delta(0)$ overstates the integral; a is the ratio of the weighted mean of δ to its value at the origin, a property of the architecture class rather than of the network, and is fitted once. Second, the k3v3 chain, which carries the third cumulants, has a smaller a and a smaller explained fraction: its defect is of higher order and less self-similar in y .

The directional second differences tell the opposite story. For a unit vector v let $\partial_v^2 e$ be the second difference of e along v at 0 and $x_v = \partial_v^2 e - \Delta e/n$ its excess over the isotropic share. On the same network the singular values of the matrix $(x_{e_i})_{i \leq n}$ decay slowly (top share 5%, top four 17%), the leading direction is orthogonal to the coherent direction $W_1^\top \mathbf{1}$ ($|\langle u_1, v_{\text{coh}} \rangle| < 10^{-3}$), and the correlation between x_v and the error is 0.18, 0.12, 0.08 (coherent, random, leading) for the Gaussian closure and 0.26, -0.05 , -0.15 for k3v3. No input direction carries the error.

3.2 The defect is a small residual of an $O(1)$ identity

At $n = 1024$, $L = 16$ (official network 0) the random-probe estimate of the Laplacian, $\Delta e \approx \frac{1}{S} \sum_s \partial_{\xi_s}^2 e$ with $\xi_s \sim N(0, I)$ (Hutchinson), fails outright: with $S = 16$ probes per block the fitted coefficient is $a \approx 0.01$ and the correlation with the error changes sign from block to block ($+0.43, -0.09, -0.44, +0.14$). The reason is quantitative. Euler–Stein says $\Delta u = u$ for the exact answer, and the chain nearly obeys it: $\Delta e(0) = e(0) - \delta(0)$ with $|\delta| \approx 10^{-2}$ against $|e| \approx 0.4$. The quantity we need, δ , is a 2% residual of an $O(1)$ identity, and the error is a tenth of that residual, so the Laplacian must be computed to a relative precision of 10^{-4} . For a unit direction v one measures $|\partial_v^2 e|_{\text{rms}} \approx 4 \cdot 10^{-3}$ at $n = 256$; since $\mathbb{E}_v (\partial_v^2 e)^2 - (\text{tr } H/n)^2 = 2\|H\|_F^2/n^2$ for v uniform on the sphere, the Hessian $H = \nabla^2 e$ of one output neuron has $\|H\|_F \approx n|\partial_v^2 e|/\sqrt{2} \approx 0.8$, comparable with $\text{tr } H \approx 0.4$. The variance of one Hutchinson probe is $2\|H\|_F^2$, so the relative noise of an S -probe estimate of $\text{tr } H$ is $\sqrt{2/S} \|H\|_F / \text{tr } H \approx 2.8/\sqrt{S}$, and the required 10^{-4} needs $S \approx 10^9$ probes. Only the exact coordinate sum, $2n + 1$ chain evaluations, resolves δ .

Proposition 3.1 (Capacity of input tilts). *Let E be a chain exact on point masses and homogeneous as in Corollary 1.3, and let H be the Hessian of e at 0 for one output neuron. (i) The error is the trace part: $u - e(0) = -\int w(\tau) \mathbb{E}_y \delta(y) d\tau$ with $\delta(0) = e(0) - \text{tr } H$. (ii) A localization that drives k orthonormal directions $v_1, \dots, v_k$ (directional tilts, or pinning of k coordinates) has drift $-\sum_{s \leq k} (\frac{1}{2} \partial_{v_s}^2 E - \partial_{\Sigma_{v_s v_s}} E)$. Over a random choice of the k directions its expectation is (k/n) times the isotropic drift, whose error-carrying part is $(k/n) \delta$, while its fluctuation over the choice is of order $\sqrt{k} \|A\|_F/n$ with $A = \frac{1}{2} H - \partial_{\Sigma} E$. The signal-to-noise ratio of a k -direction scheme is*

therefore $\sqrt{k}|\delta|/\|A\|_F$. (iii) With $|\delta|/\|A\|_F \approx 10^{-2}$ as measured, a k -direction scheme resolves the error only for $k \gtrsim 10^4 \gg n$, and the cheapest exact scheme on the input side is the coordinate sum, $2n+1$ chain evaluations, i.e. $2n+1$ pair chains at $n=1024$, far outside any budget. Input localization is useless for estimation; it is a diagnostic.

Proof. (i) is Corollary 1.3. (ii): the directional drift of Corollary 1.4 is a quadratic form in v with matrix $\frac{1}{2}H - \partial_\Sigma E$, whose average over the Haar measure on k -frames is k/n of its trace, and whose variance over the frame is $\sum_s \text{Var}(v_s^\top A v_s) \approx 2k\|A\|_F^2/n^2$ for A the matrix of the form. (iii) follows by comparing the two. $\square$

This is the negative half of the stage-8 dictionary: the input is the *commutative* side, its tilts are the modular tilts of the Gaussian, and the chain's defect is invisible to any finite family of them because it is a trace. The positive half is that the trace can be computed *inside* the chain, as the second-order response of the closure to a Gaussian shift of the mean, which is the subject of Sections 2 and 4: there the directions are neurons, the tilt responses are the births at the kinks, and the sum over directions is a sum over neurons that the chain already performs.

4 The collective mode is the central variable

The stage-8 principle splits a localization into its trivial (central) and Cartan parts. For the MLP the central variable is the collective activity of a layer: the direction $s = \mathbb{E}[z^2]$ in pre-activation space, which is also the top eigenvector of the covariance once depth has built it up. This section proves two small exact statements about it and reports what the chain does to it.

4.1 A scale mixture is repaired exactly at first order by the (2,2) channel

Lemma 4.1 (Scale mixture). *Let $z \mid \varepsilon \sim N(m, (1+\varepsilon)^2 S)$ with $\mathbb{E}\varepsilon = 0$, $\mathbb{E}\varepsilon^2 = v$ small. Then the cumulants of z are those of $N(m, (1+v)S)$ at second order and $\kappa_4(z_a, z_b, z_c, z_d) = 4v(S_{ab}S_{cd} + S_{ac}S_{bd} + S_{ad}S_{bc}) + O(v^2)$, a pure (2,2)-type tensor. The Gaussian closure applied to the true covariance $(1+v)S$ gives $\mathbb{E} \text{relu } z_a$ with error $-\frac{v}{2}\sigma_a(\alpha_a^2 - 1)\varphi(\alpha_a) + O(v^2)$, and the κ_4 term of the mean readout (10) with $\kappa_4(z_a) = 12v\sigma_a^4$ is $+\frac{v}{2}\sigma_a(\alpha_a^2 - 1)\varphi(\alpha_a)$: the two cancel.*

Proof. $\mathbb{E}[(1+\varepsilon)^4] = 1 + 6v + O(v^2)$ and $\mathbb{E}[(1+\varepsilon)^2]^2 = 1 + 2v + v^2$ give the fourth cumulant. Put $f(\varepsilon) = (1+\varepsilon)\sigma d_0(\alpha/(1+\varepsilon))$; the true mean is $\mathbb{E}f(\varepsilon) = f(0) + \frac{v}{2}f''(0)$ and the closure's is $f(\varepsilon^*)$ with $(1+\varepsilon^*)^2 = 1+v$, i.e. $f(0) + \frac{v}{2}f''(0)$. From Lemma 2.1, $f'(0) = \sigma[d_0 - \alpha d_1] = \sigma\varphi$ and $f''(0) = \sigma\alpha^2\varphi$. $\square$

So the (2,2) channel of the chain (k22: $\kappa_4(h_a, h_a, h_b, h_b)$ born exactly from the Hermite series of $\text{Cov}((h_a - \mu_a)^2, (h_b - \mu_b)^2)$, transported by $(W \circ W)K(W \circ W)^\top$ with the gate $tt^\top$ and the diagonal class removed once) is the first-order correction for a fluctuating collective scale. Its gain projection $4v = s^\top K_{22}s/(s^\top s)^2$ measures v directly.

4.2 The sensitivity law

Proposition 4.2 (Sensitivity). *A relative error ε_a in the pre-activation variance of neuron a changes its post-activation mean by $\frac{1}{2}\varepsilon_a\sigma_a\varphi(\alpha_a)$, and an error $\delta\lambda$ in the top eigenvalue of the covariance of layer l (eigenvector u) changes the pre-activation variance of every neuron b of layer $l+1$ by $\delta\lambda(w_b \cdot u)^2 \approx 2\delta\lambda/n$. On the official networks ($\sigma \approx 2.8$, $\varphi(\alpha) \approx 0.3$, $\text{var} \approx 8$, $\lambda_1 \approx 12$ at depth) a uniform relative variance error of 10^{-3} moves every mean by $4 \cdot 10^{-4}$, seven times the*

rms error of the chain, while a 2% error of the top eigenvalue moves the next layer's variances by $5 \cdot 10^{-5}$ relative and the means by $2 \cdot 10^{-5}$ per layer.

Proof. $\partial_{\sigma^2}[\sigma d_0(m/\sigma)] = \frac{1}{2\sigma}(d_0 - \alpha d_1) = \varphi/(2\sigma)$, multiplied by $\varepsilon\sigma^2$; and $w_b^\top(\delta\lambda uu^\top)w_b$ with $w_b \cdot u \sim N(0, 2/n)$ for He rows. $\square$

The bulk of the covariance may therefore be wrong by percents without consequence (it enters the next layer's variances through $\sum_c S_{bc}^2$ -type sums that average out), while the diagonal must be right to 10^{-4} and the collective mode to 10^{-3} : that is what makes the central variable the one that matters.

4.3 Measurements on the official network 0

From the Monte Carlo cache (2^{20} inputs; `scripts/diag-eig-v3.py`, `diag-onestep.py`):

- The collective mode emerges with depth: the top four eigenvalues of the pre-activation covariance are 8.08, 7.89, 7.77, 7.71 at layer 0 and 16.3, 5.36, 4.88, 4.62 at layer 15.
- The chain's trace is exact to 10^{-3} at every layer and its diagonal to 10^{-4} ; the bulk (the complement of the top four modes, off-diagonal part) is off by 2–4%; the top eigenvalue is correct to $2 \cdot 10^{-3}$ through layer 5 and then 1.0, 1.2, 1.5, 1.8, 1.9, 1.8, 1.7, 1.8, 1.5, 1.6% *low* at layers 6–15, with the variance along s equally low. By Proposition 4.2 this is a $2 \cdot 10^{-5}$ per layer effect on the means, a tenth of the measured error: the top mode is the *symptom* of the deficit, not its carrier.
- One-step closure from Monte Carlo inputs at layers 11, 13, 14: the Gaussian closure loses 3.2, 1.5, 1.1% of the top mode in one layer; adding the Monte Carlo (2, 1) slices (the κ_3 covariance correction (9)) leaves $-0.13, -0.09, -0.09\%$; adding the (2, 2) slices changes that by less than $3 \cdot 10^{-4}$; adding the (3, 1) slices makes it worse ($-0.19, -0.15, -0.16\%$). The κ_3 slice is the whole one-step story of the collective mode, and the sign of the (3, 1) term says its first-order transport is not the right object to add.
- The chain's own transported slices at those layers have correlation 0.965 with Monte Carlo and slope 0.92, 0.90, 0.91; its (2, 2) slices have correlation 0.22–0.25 and slope 0.53–0.68, and its gain projection $4v$ is 3.6, 1.9, $1.5 \cdot 10^{-5}$ against 6.6, 4.6, $4.0 \cdot 10^{-5}$: the scale-mixture variance is $v \approx 10^{-5}$, so by Lemma 4.1 the (2, 2) channel is worth $\sim 10^{-5}$ per neuron and layer, which is why the chain's (2, 2) errors are harmless.

The error accounting that follows is then simple. A 10% deficit of the transported slice leaves 10% of the 1–3% one-step loss of the collective mode in place, 0.1–0.3% per layer, which accumulates to the 1.5–1.9% measured; the same deficit leaves 10% of the κ_3 corrections to the diagonal and the means in place at every layer, and the per-layer injected mean errors of $3 \cdot 10^{-4}$ (Section 7) are of that size: 10% of $\kappa_3(z_a)\alpha\varphi/(6\sigma^2)$ at $\kappa_3 \approx 3$, which is $3 \cdot 10^{-4}$. The chain's error is the deficit of its (2, 1) transport, in the diagonal readout and in the collective mode alike.

5 Exact first-order transport through ReLU

The chain of this note (`whest/k3chain3.py`) transports the third cumulant in the CP-leg form of [2]: each kink neuron m of layer l is a source with centre leg $P = W \cdot e_m$, arm $A = W \cdot a_m$, $a_m = \text{diag}(g) C e_m$, strength $w_m = \varphi(\alpha_m)/\sigma_m$, and the two further legs B (the Gram–Charlier arm fed by the incoming (2, 1) slices) and R (the residual that makes the slice and the diagonal exact at birth). The readouts in pre-activation space are the diagonal $\kappa_3(z_a) = D3_a$ and the slices

$D21_{ab} = \kappa_3(z_a, z_a, z_b)$. What is new relative to [2] is the *transport* of a slice through a ReLU layer to first order in the cumulants, in closed form, which we state and then measure.

Theorem 5.1 (First-order transport of the (2,1) slice). *Let $z \sim$ a law with Gaussian part $N(m, S)$ and small third and fourth cumulants, $h = \text{relu}(z)$, and write $g = \Phi(\alpha)$, $w = \varphi(\alpha)/\sigma$, $\mu_a = \sigma_a d_0(\alpha_a)$, $t_a = g_a - \mu_a w_a$, $\epsilon_a = 2\mu_a(1 - g_a)$, $c_a = g_a(1 - g_a) - \mu_a w_a$. To first order in the cumulants of z and to the orders in ρ kept by the chain,*

$$\kappa_3(h_a, h_a, h_c) = \left[g_a^2 + c_a \right] g_c \kappa_3(z_a, z_a, z_c) + \frac{1}{2} \epsilon_a w_c \kappa_3(z_a, z_c, z_c) + \frac{1}{2} t_a w_c \kappa_4(z_a, z_a, z_c, z_c) + (\text{born}), \quad (8)$$

$$\text{Cov}(h_a, h_b) = \text{Cov}^{\text{Gauss}}(h_a, h_b) + \frac{1}{2} (w_a g_b \kappa_3(z_a, z_a, z_b) + g_a w_b \kappa_3(z_a, z_b, z_b)) + \frac{1}{4} w_a w_b \kappa_4(z_a, z_a, z_b, z_b), \quad (9)$$

$$\mathbb{E}[h_a] = \sigma_a d_0(\alpha_a) - \frac{\alpha_a \varphi(\alpha_a)}{6\sigma_a^2} \kappa_3(z_a) + \frac{(\alpha_a^2 - 1) \varphi(\alpha_a)}{24\sigma_a^3} \kappa_4(z_a), \quad (10)$$

where “born” is the exact κ_3 of relu of the Gaussian part (all Hermite orders of the slice, `relu2_hermite`, and the exact diagonal `k3_exact`), and the gate of the transported slice, $g_a^2 + c_a = g_a - \mu_a w_a$, is the exact coefficient $\mathbb{E}[\theta_a \partial_a (\text{relu } z_a - \mu_a)^2]/2$ rather than the Gaussian g_a^2 .

Proof sketch. Expand the law of z as a Gram–Charlier perturbation of its Gaussian part, $\nu = \nu_G(1 + \frac{1}{6}\kappa_3 : \text{He}_3 + \frac{1}{24}\kappa_4 : \text{He}_4)$, and compute $\mathbb{E}[(h_a - \mu_a)^2(h_c - \mu_c)]$ by integrating the Hermite polynomials against the ReLU moments: Stein’s lemma turns each He_3 against a product of ReLUs into third derivatives, which are sums of gate products $\theta_a \theta_a \theta_c$ (the first term), of $\theta_a \delta_c \text{relu}_c$ -type terms, i.e. a birth w_c at c against the transposed slice (the second term), and the He_4 term gives the (2,2) feed with the single birth w_c (third term). The coefficients t_a, ϵ_a, c_a are the exact one-dimensional integrals $\mathbb{E}[\theta_a (\text{relu } z_a - \mu_a) \cdot]$ listed in the code. The covariance and mean formulas are the same computation one order lower. $\square$

The three transported terms are what the chain calls the gated slice (`gate_res`), the transposed-slice feed (`feedT`) and the $\kappa_4 \rightarrow \kappa_3$ feed (`feedK22`). Their effect on the official network 0 is the ladder of Section 7: the transported slices rise from 75% to 90–92% of their Monte Carlo values at depth and the final MSE falls from $8.5 \cdot 10^{-7}$ to $5.7 \cdot 10^{-7}$.

What remains open in the transport. The first term of (8) transports the slice with the gate of the pair (a, c) taken at $\rho_{ac} = 0$ plus the chain’s ρ -expansion; the exact gate $\mathbb{E}[\theta_a \theta_a \theta_c(\cdot)]$ at the true correlation differs at order ρ_{ac} , which is small per entry but coherent along the collective mode. Section 8 lists the ρ -exact gate as the first open item, since the 8–10% deficit of the transported slice at depth is precisely of this size.

6 What the chain costs: the pair floor

The score multiplies the error by $\max(0.1, C/B)$ with C the metered count of the grader’s own numpy wrapper (`flopscope 0.12.1`) and $B = 2^{41}$. We read the wrapper’s rules off its source and measured the bills of the chain’s building blocks at $n = 1024$ (`notes/compute/FLOPSCOPE.md`); the facts that decide a design are these.

Rules. Every call bills $\lfloor \text{textbook count} \times \text{dtype rate} \times \text{weight} \rfloor$. The count is the textbook one, not what BLAS does: a `matmul` $(m, k)(k, n)$ is $2mkn - mn$ whatever algorithm the library uses, so a Strassen level written out of seven smaller `matmul` calls *is* billed at 7/8. The dtype rate is 1 for float32 and 2 for float64. The weight is 0 for views, transposes, `zeros`; 1 for arithmetic, reductions, contractions, copies, `sqrt`, `stats.norm.cdf/pdf` (whose counts are 48 and 27 per element, computed in float64), `polyval` ($2d$ per element) and all of `linalg`; 4 for `where`, gathers and sorts; 16 for `exp`, `log` and every random sampler. A contraction whose output the engine can prove symmetric is billed on the unique entries: the sandwich $WCW^\top$ written as one `einsum('ij,jk,lk->il', W, C_sym, W)` with the same array W in both outer slots and C tagged `as_symmetric` bills $3n^3$ in float32 ($6n^3$ in float64) against $4n^3$ ($8n^3$) for two `matmul`s; the saving is lost if the two products are separate calls. `eigh` is $9n^3$, `cholesky` $n^3/3$, `inv` $2n^3$, `solve` $\frac{2}{3}n^3 + 2n^2r$.

Budget. The floor $0.1B = 2.2 \cdot 10^{11}$ equals $205n^3$ over the 16 layers, i.e. $12.8n^3$ per layer: six float32 (n, n, n) `matmul`s, or four float32 symmetric sandwiches, or three float64 `matmul`s. Nothing is gained below the floor, so a chain should spend exactly it. The Gaussian pair chain, one sandwich per layer, costs $0.024B$ in float32 (one quarter of the floor) and $0.047B$ in float64. Elementwise work on $n \times n$ arrays is negligible unless repeated dozens of times per layer: the Hermite series of the pair moment to order K is K Hadamard products, $2Kn^2$, i.e. $0.002n^3$ per term.

Proposition 6.1 (Pair floor). *Any estimator that forms the pre-activation covariance of every layer from that of the previous one pays at least one sandwich per layer: $3n^3$ in float32 with the symmetric engine, $L \cdot 3n^3 = 0.024B$ at $n = 1024$, $L = 16$. Every further $n \times n$ object transported through W (a κ_3 slice, a $(2, 2)$ slice, a gate-correlation matrix) costs another $2n^3$ to $3n^3$ per layer, i.e. $0.016B$ to $0.024B$ over the network. Under the floor a chain may therefore carry the pair plus about four such objects per layer; the leaders' bills ($0.105B$, $0.144B$) correspond to five to seven float32 sandwich-equivalents per layer, and the open-source third-cumulant chain at $0.25B$ to a float64 implementation of the same count.*

Naive counts mislead. The count our chain returns (FMA = 2, no dtype rate, no symmetry) overstates the float32 symmetric bill by $2.7\times$ and understates a float64 plain-matmul bill by $2\times$. The *pair floor* and the *per-object price* above are the only two numbers a design needs; every proposal in Section 8 is costed in units of “ $n \times n$ objects transported per layer”.

Precision. float32 arithmetic (relative 10^{-7} per operation) propagates the covariance through sixteen layers of 1024-term sums to a relative 10^{-6} , far below the 10^{-4} rms of a 10^{-8} -MSE estimator; but the quantities that are small differences of $O(1)$ numbers (the Euler–Stein defect $e - \Delta e$, differences of nearly equal slices) must be formed in float64 or arranged to avoid cancellation.

6.1 The memory of the third cumulant and the transport cocycle

The organisers' chain keeps the all-distinct part of κ_3 in factored form $\text{Sym} \sum_r A_{ir} B_{jr} C_{kr}$ and adds $2n$ columns to each leg at every layer (the star block of the births and the residual block that makes the $(2, 1)$ and (3) slices exact), so the rank at layer l is $2ln$ and the transport and readout cost $18ln^3$ per layer, $1.2B$ in float32 over the network: 93% of the reference's own count. The theory says where to cut. A block born at layer l' is, at layer l , the image of its birth legs under

the gated transport cocycle

$$T_{l \leftarrow l'} = W_l \text{diag}(g_{l-1}) W_{l-1} \text{diag}(g_{l-2}) \cdots W_{l'+1} \text{diag}(g_{l'}),$$

a product of $l - l'$ independent gated random matrices. Such products contract onto their leading Oseledets directions, and the leading directions are the collective mode of Section 4: in the stage-8 language the old sources live on the leaf space, and the shared basis of their transport is the transverse measure’s support. On network 0 (`scripts/diag_cocycle.py`) the effective rank $\|T\|_F^2/s_1^2$ of $T_{l \leftarrow l-a}$ is 164, 92, 65, 47, 29, 23 at ages $a = 1, 2, 3, 4, 6, 8$ (independent of l to a few percent), and a random block transported for a ages has 51, 70, 81, 87, 94, 96% of its energy in the top 128 left singular directions and 77, 92, 95, 98.5, 99.7, 99.8% in the top 256. Sources of age ≥ 2 can therefore be merged into a *Tucker tier* $\text{Sym}((Q \otimes Q \otimes Q)S)$ with one orthonormal basis $Q \in \mathbb{R}^{n \times k}$ shared by the three legs and a symmetric core $S \in \mathbb{R}^{k \times k \times k}$ (`whest/kprop3c.py`): the basis is transported as $W \text{diag}(g)Q$ and re-orthonormalised, the core rotated by the $k \times k$ factor, and an aged block is folded in by refitting the basis to the dominant directions of $[Q \text{diag}(\nu), A, B, C]$ and projecting. Its cost per layer is $2nk^3 + 4n^2k$ for the two slice readouts and $2nk^3$ for a merge, i.e. $19n^3$ at $k = 128$ with no young tier, against $18ln^3$ dense; at $k = 256$ the cubic core ($2nk^3 = 32n^3$ per readout) is already too expensive, which is the statement that a generic order-three tensor on a k -dimensional leaf space has $k^3/6$ degrees of freedom and cannot be read for less. The young tier (ages $< w$) costs $16n^3$ per source-age. Table 2 gives the resulting bills under the grader’s rules; the measured accuracy of each configuration is in Section 7.

(w, k)	n^3/layer	C/B (float32)	(w, k)	n^3/layer	C/B (float32)
(0, 128)	19	0.148	(2, 128)	51	0.398 (0.32 with Strassen)
(1, 128)	35	0.273 (0.23)	(3, 128)	67	0.523 (0.41)
(0, 256)	92	0.716	dense reference	$18l$	1.195

Table 2: Billed cost of the two-tier chain (pair chain, metric, young tier, Tucker tier, merge, elementwise terms).

6.2 Why the memory resists Frobenius compression

The measured accuracy of the two-tier chain (Section 7) is far worse than the energy fractions of the cocycle suggest, and the reason is structural. The slices are read through *Hadamard* products of the legs, $(A \circ B)C^\top$: for a residual block transported for a ages the $(2, 1)$ slice is $\frac{1}{3}[2((TR) \circ T)T^\top + (T \circ T)(TR)^\top]$ with $T = T_{l \leftarrow l-a}$. Projecting a leg on the top- k subspace of T controls $\|T - QQ^\top T\|_F$, but the Hadamard square $(QQ^\top T) \circ (QQ^\top T)$ is not the projection of $T \circ T$: the square of a rank- k matrix has rank up to $k(k+1)/2$ and lives in a different subspace. A leg truncation that loses a fraction ϵ^2 of the energy therefore loses a fraction $\sim 2\epsilon$ of the slice, and by the sensitivity law of Proposition 4.2 the slices of the sources that matter must be right to 10^{-3} : the leg energy must be captured to 10^{-6} , which the cocycle spectrum ($s_{256}/s_1 = 0.5, 0.3, 0.18, 0.17$ at ages 1–4) is nowhere near. Compression of the kind that works for a *linear* transport (keep the dominant directions of T) does not work for a cubic readout, except for sources old enough that their whole contribution is small: at age ≥ 4 the sources are worth a factor 6 in MSE, so a 5% slice error on them costs $\sim 10\%$, which is the regime in which the open-source chain’s shared basis ($k = 384$, window 4) loses its reported 3.5%. The arithmetic of the pair floor then says that an exact-first-order chain cannot be brought near $0.1B$ by compressing its memory: the young

window alone, four ages at $16\text{--}24n^3$ each, exceeds the whole budget. What a competitive chain at $0.1B$ must do instead is keep *fewer* ages and recover the systematic part of what is lost by calibration on the public networks; Section 7 measures how much of the loss is systematic (a scale of the old sources’ slices) and how much is shape.

7 Measurements on the official networks

All numbers are for the official network 0 ($n = 1024$, $L = 16$, reference means from 10^9 Monte Carlo inputs) unless stated; network 1 is used as the held-out check where available.

7.1 The error ladder of the clean chain

estimator	final-layer MSE	what is added
Gaussian closure (gauss_chain)	$4.06 \cdot 10^{-6}$	—
κ_3 CP legs, stars only, no hub	$3.76 \cdot 10^{-6}$	births w , arms d
+ hub: $D21 \rightarrow$ covariance (9), GC arm	$8.37 \cdot 10^{-7}$	transported
+ exact birth slice and diagonal (residual leg R), exact gates	$8.52 \cdot 10^{-7}$	Theorem 5.1, first
+ transposed-slice feed, $\kappa_4 \rightarrow \kappa_3$ feed, (2, 2) channel, exact diagonal κ_3	$5.74 \cdot 10^{-7}$	Theorem 5.1, second
Gaussian closure + Euler–Stein defect, $e - a\delta(0)$, $a = 0.535$	$4.35 \cdot 10^{-7}$	Corollary 1.3, exact δ
Gaussian closure + midpoint $(e + \Delta e)/2$ (no fitted constant)	$4.50 \cdot 10^{-7}$	same
organisers’ chain, $k_{\max} = 3$ simple, numpy port (kprop3), radial κ_4 off	$5.63 \cdot 10^{-7}$	exact first-order κ_3 trans
same, radial κ_4 on (one scalar per layer)	$3.53 \cdot 10^{-8}$	the central variable
open-source third-cumulant chain (504aldo), float32, 33 fitted scalars, $0.25B$	$2.13 \cdot 10^{-8}$	
exact first-order $K = 3$ reference of the organisers (no κ_4 , no fits)	$4.3 \cdot 10^{-8}$	
leaders (raw)	$1.1\text{--}1.5 \cdot 10^{-8}$	

The decisive line is the pair in the middle: the organisers’ exact first-order chain, ported to numpy and matched to the torch reference bit for bit, reaches $5.6 \cdot 10^{-7}$ on network 0 when its fourth cumulant is dropped, the accuracy of our own chain, and $3.5 \cdot 10^{-8}$ when the single radial scalar $c = \frac{3}{n(n+2)}(\sum_{a \neq b} \kappa_4(h_a, h_a, h_b, h_b) + \sum_a \kappa_4(h_a))$ per layer is kept. One number per layer, the collective scale of Lemma 4.1, is worth a factor 16 in MSE; our chain’s transported (2, 2) slice, which carries n^2 numbers and is harmful off the diagonal, was worth nothing. The reference’s own naive count is $3.4 \cdot 10^{12}$ FLOPs ($1.5B$ in float32), 93% of it in the transport and readout of the factored κ_3 whose rank grows by $2n$ per layer; the cost problem is the memory of the third cumulant, not the fourth.

Two readings. First, the Euler–Stein defect of the *Gaussian* closure, with no cumulant machinery at all, reaches the accuracy of our third-cumulant chain: on network 0 the exact $\delta(0)$ (coordinate Laplacian, 2049 chain evaluations on Modal) has correlation 0.931 with the error, explains 89% of its mean square, and the fitted $a = 0.535$ is close to the parameter-free midpoint $a = \frac{1}{2}$. Theorem 1.2 is exact; the 11% not explained is the variation of $\delta(y)$ over the Eldan path. Second, the organisers’ exact first-order $K = 3$ reference is $13\times$ better than our chain and carries no fitted constant, so the gap is in the first-order transport, not in calibration or in κ_4 .

7.2 Where the chain’s error is injected

With $e_l = \text{out}_l - u_l$ and the injected part $e_l - \Phi_l \cdot (W_l e_{l-1})$ (the error not explained by the linear propagation of the previous layer’s error through the gates), **scripts/injected.py** gives rms

injected errors of 2.0, 1.1, 1.6, 2.1, 2.3, 2.5, 2.9, 2.7, 2.8, 2.9, 2.8, 2.7, 2.8, 2.9, $3.0 \cdot 10^{-4}$ at layers 1–15 against a total rms error growing from $2.0 \cdot 10^{-4}$ to $7.6 \cdot 10^{-4}$: the chain injects a nearly constant $3 \cdot 10^{-4}$ per layer from layer 4 on, and the injected error has no coherent component along the mean direction (share $< 10^{-2}$ beyond layer 2). This is the signature of a *per-neuron*, incoherent deficit in the transported slices, consistent with Section 4: the transported $D21$ has correlation 0.965 with Monte Carlo, i.e. an incoherent relative error of $\sqrt{1 - 0.965^2} \approx 26\%$ on top of the 8–10% coherent deficit, and by the empirical law of [2]’s reimplementation (extra $\text{MSE} \approx 4 \cdot 10^{-6} \varepsilon^2$ for a relative rms slice error ε) a slice error of 35% accounts for the whole $5.7 \cdot 10^{-7}$.

7.3 What the open-source chain does that ours does not

A reading of the open-source chain’s derivation notes against `k3chain3.py` (delegated, cross-checked on the formulas) locates the difference in exactly the first-order transport: (i) its slice update carries every elementwise diagram of first order in the cumulants, i.e. the ρ -dependence of the gates ($D21_{ac} C_{ac}^k \times$ Wick coefficients) and the $\kappa_3(z_a)$ -driven terms, where ours keeps the gates at $\rho = 0$; each such term is $O(\rho) \approx 3\%$ per layer and compounds over fifteen transports; (ii) its fourth cumulant is the exact diagonal plus an isotropic off-diagonal $\lambda_l C^{\text{off}}$ with one fitted λ_l per layer, and it *feeds* κ_4 into the fully off-diagonal κ_3 (the largest single term beyond the base chain, $2\times$ in MSE), whereas our transported $(2, 2)$ slice is measured by them to be harmful off the diagonal ($1.47\times$ worse than isotropic); (iii) the feedback coefficients of the GC arm are $(\frac{1}{2}, \frac{1}{2})$ by the first-order Edgeworth count against our $(1, \frac{1}{2})$; on network 0 changing ours to $\frac{1}{2}$ moves the MSE from 5.74 to $6.04 \cdot 10^{-7}$, so this is not the bottleneck. The ρ -exact gates of (i) are the closed forms of Section 2: by Lemma 2.1 the coefficient of $D21_{ac}$ in $\kappa_3(h_a, h_a, h_c)$ is $G_{ac} - \mu_a w_a \Phi((\alpha_c - \rho_{ac} \alpha_a) / \sqrt{1 - \rho_{ac}^2})$ with $G_{ac} = \mathbb{E}[\theta_a \theta_c]$ the gate covariance (5), which reduces to $t_a \Phi_c$ at $\rho = 0$, and likewise for the transposed slice and the $(2, 2)$ feed with conditional means and densities at $z_c = 0$. Every coefficient of the first-order transport is a Hermite series obtained by index shifts, which is how a system built on this note generates its term table (Section 8).

8 Dictionary and open problems

8.1 Dictionary

stage 8 (commutant, modular tilts, leaf space)	stage 9 (MLP mean estimation)
state ν on the algebra; affine pairing $\nu(F)$	input law; the exact answer $u(\nu) = \nu(h_L)$
localization process (positive martingale in the commutant)	Eldan tilt of the input, directional tilt, pinning; on the neuron side, the layer-by-layer closure
driving covariance C_t ; trickle-down term KC_tK	which errors a scheme sees: the second variation E'' against $d\langle\nu\rangle$ (Theorem 1.2)
modular tilt of a Gaussian state	shift of the mean; its action on the chaos expansion is the index shift (Lemma 2.1)
central (trivial) part of the localization	the collective activity mode $s = \mathbb{E}z^2$, a scale mixture (Lemma 4.1); the transverse measure is its law
Cartan part	the bulk of the covariance, harmless at percent accuracy (Proposition 4.2)
Kikuchi–Toeplitz quantization	the heat semigroup in the input mean (Price’s theorem); the exact answer is the Berezin transform of the network function, the chain a Toeplitz symbol that is not heat-harmonic, and the anomaly is the Euler–Stein defect
capacity of a localization	Proposition 3.1: input tilts have signal-to-noise $\sqrt{k} \delta /\ A\ _F \approx 10^{-2}\sqrt{k}$
births / creation operators	the kink births $w_a = \varphi(\alpha_a)/\sigma_a$, second derivatives of the tilt response ((3))

8.2 What the theory prescribes for a system, and what is open

The decisive constraints are the pair floor of Proposition 6.1 (four $n \times n$ objects per layer besides the covariance, at the $0.1B$ floor) and the error accounting of Section 4 (the $(2, 1)$ transport carries the error; the $(2, 2)$ channel is worth 10^{-5} ; the $(3, 1)$ term has the wrong sign at first order). A system built on this note therefore spends its budget on *one* object, the $(2, 1)$ slice, transported with the ρ -exact gate, in float32 with the symmetric engine, and corrects the rest by calibration fitted on the public networks, in this order of expected value:

1. **ρ -exact slice gate.** Replace the gate $g_a - \mu_a w_a$ of (8) by its value at the true correlation ρ_{ac} (one more Hermite series in ρ , $O(Kn^2)$, no new n^3 object). The 8–10% slice deficit is of the size of the $O(\rho)$ correction along the collective mode.
2. **Euler–Stein merge.** $e - a\delta(0)$ with a fitted once: a factor 3 (k3v3) to 15 (Gaussian closure) at $n = 256$; at $n = 1024$ the exact δ of the Gaussian closure is measured in Section 7. Within budget only through the closable part of the second-order response of Theorem 2.2, i.e. the covariances $\mathbb{E}[\delta m \delta m^\top]$, $\mathbb{E}[\delta m_a \delta S_{ab}]$, $\mathbb{E}[\delta m_a \delta S_{bb}]$, $\mathbb{E}[\delta S_{aa} \delta S_{bb}]$, $\mathbb{E}[\delta S_{ab} \delta S_{aa}]$, which are the same five objects a κ_3 chain carries under other names; whether the closable part correlates with the exact δ as well as δ correlates with the error is the open experiment of Section 7.
3. **Per-layer slice scale.** A fitted multiplier $1 + \gamma_l$ on the transported slices (the chain’s d3scale generalised per layer): one number per layer, fitted on the public networks, directly targets the measured deficit.
4. **Noncommutative version.** The gates $\theta(z_a)$ of one layer generate a commutative algebra; across layers the gate algebras do not commute with the weights, and the chain’s closure is a product state on the gate algebra of layer l conditioned on the surrogate of layer $l - 1$. The stage-8 KMS condition for the modular flow of the Gaussian state, restricted to the gate algebra, is the heat equation of Section 2; its failure across layers is the defect. Writing the

chain as a state on the free product of the gate algebras, with the trickle-down of [3] as its Cartan expansion, is the theoretical continuation.

A Verification

`scripts/verify_stage9.py` checks, in order: (1) Lemma 2.1 to $2 \cdot 10^{-8}$ for $k \leq 10$; (2) Price’s theorem and Stein’s lemma for the bivariate RELU moment, $\partial_{m_a} M_{ab} = \mathbb{E}[\theta_a \text{relu } z_b]$, $\partial_{S_{ab}} M_{ab} = \mathbb{E}[\theta_a \theta_b] = \partial_{m_a} \partial_{m_b} M_{ab}$, against $4 \cdot 10^6$ Monte Carlo samples; (3) Theorem 2.2 on a network with $n = 6$, $L = 3$: the gradients of the first-layer moments in closed form against finite differences ($1.6 \cdot 10^{-7}$ on a scale of 2), and $\delta(0)$ by finite differences in the input mean against $-\langle \nabla^2 \mathcal{G}, \Gamma \rangle$ with the Hessian of the downstream chain by finite differences in the 27 moment coordinates: agreement to $7 \cdot 10^{-7}$ on $|\delta|_{\text{rms}} = 0.28$; (4) Lemma 4.1 by Gauss–Hermite quadrature in the scale, $v = 10^{-2}$: the closure error matches $-\frac{v}{2}\sigma(\alpha^2 - 1)\varphi$ and the κ_4 readout cancels it to the $O(v^2)$ level for $\alpha \in \{-1, 0, 0.7, 1.5\}$. `scripts/defect_small.py`, `lap_exact.py`, `lapx_merge.py` produce the numbers of Sections 3 and 7; `diag_eig.v3.py`, `diag_onestep.py`, `injected.py` those of Section 4.

References

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